

Ecological niches and geographic distributions, a perspective from Macroecology

Modeling the distribution of marine biodiversity under global climate change

Macroecology: the big picture

A statistical approach **focusing on large-scale ecological patterns** (not a separate field of Ecology; coined ~30 years ago).

Explores **relationships between biodiversity patterns** (e.g., species abundance levels and the distribution of genes, species, ecosystems) **and the environment**.

Seeks general principles governing ecological systems, often ignoring fine-scaled details.

e.g.,

What environmental factors drive the distribution of a mangrove species?

How global environmental changes may impact kelp forests biodiversity?

Macroecology: the big picture

Compares **populations** (in different locations), **species** or **ecosystems**, across large areas, long time periods, **to generalize and scale up results to global processes** - to figure out the general distribution rules or patterns.

In opposition to reductionist and localized experimental approaches, which have difficulty in generating predictive theory.

The non-randomness of life

Species aren't found everywhere – their distributions are limited.

Macroecology helps understand these limits, which are mostly set by:

Niche requirements: environmental tolerances and resource needs.

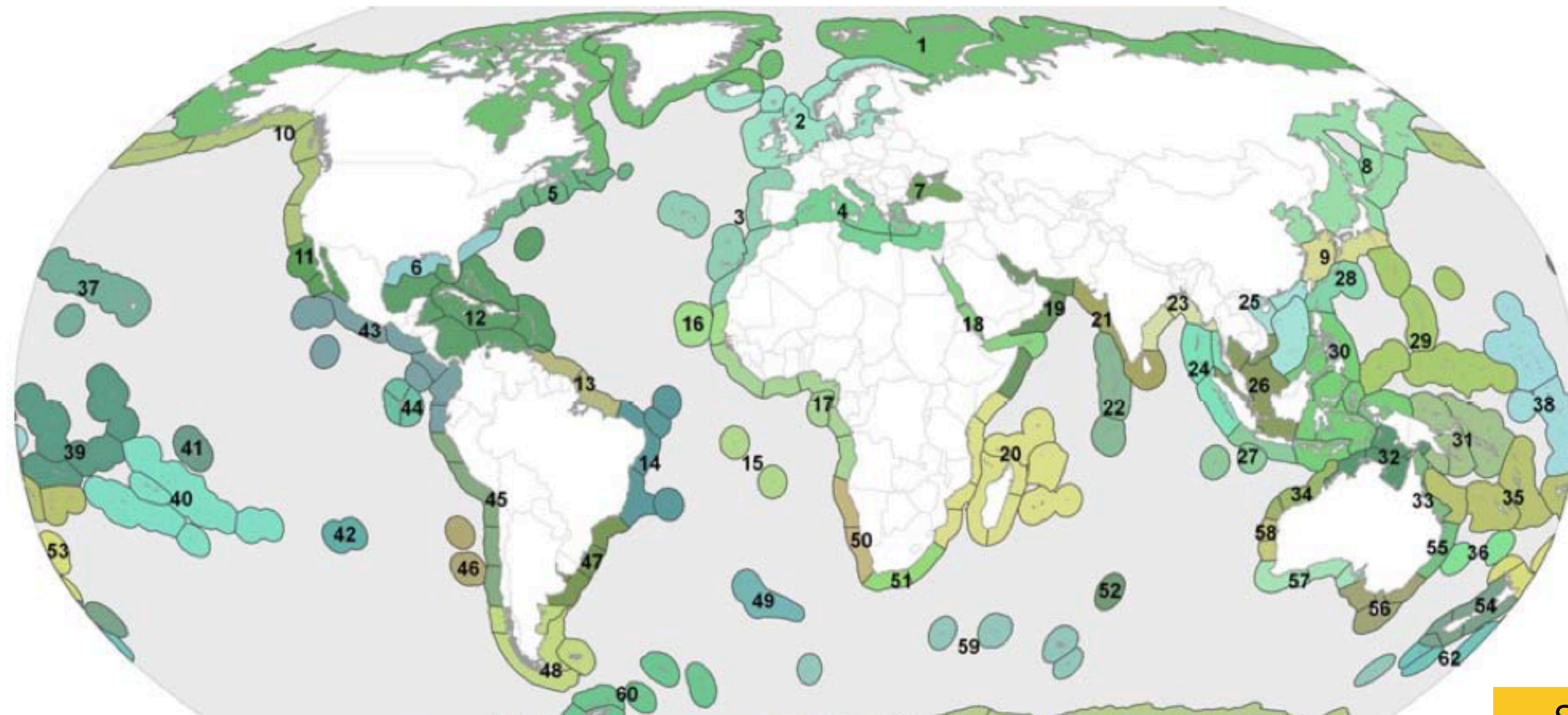
Dispersal constraints: ability to reach and establish in suitable areas.

Without these constraints, species could potentially occur everywhere, leading to absent or random spatial patterns in distribution and diversity.

Well defined biogeographic patterns

Multiple classification systems based on **distinct species assemblages and levels of endemism** propose sharp biogeographic regions - within / between regions, similar / dissimilar biodiversity.

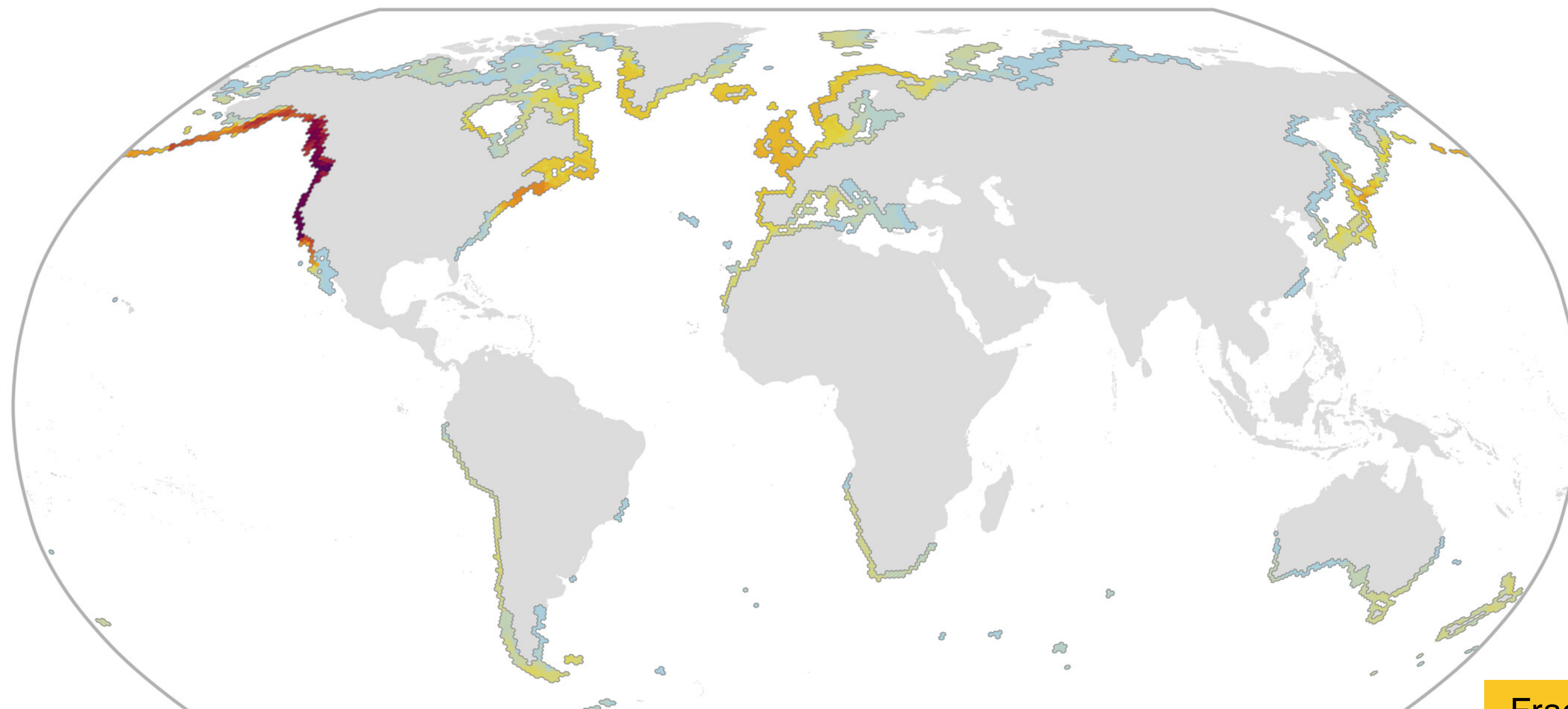
e.g., Marine Ecoregions of the World; nested system (12 realms, 62 provinces, and 232 ecoregions).



Well defined biogeographic patterns

Multiple **estimates of global species richness and endemism** for different groups identify **regions with higher and unique biodiversity**.

e.g., Global biodiversity patterns of marine forests of brown macroalgae.



The ecological niche: a core concept

A foundational concept in Macroecology. Evolved through different perspectives:

Grinnellian niche (Joseph Grinnell, 1924): the abiotic conditions (e.g., temperature tolerance) and habitat requirements (e.g., substratum type) that determine where a species can survive and reproduce.

Eltonian niche (Charles Elton, 1927): the species' functional role within its community, in particular the interactions it has with other organisms (e.g., competition, predation).

Hutchinsonian niche (G. Evelyn Hutchinson, 1957): a "n-dimensional hypervolume" that includes the **abiotic** conditions (i.e., environmental), the **biotic** interactions (e.g., availability of prey) and **resource** requirements (e.g., availability of nursing sites) under which **a species can maintain viable populations** (i.e., survival and reproduction).

Fundamental vs. realized niche

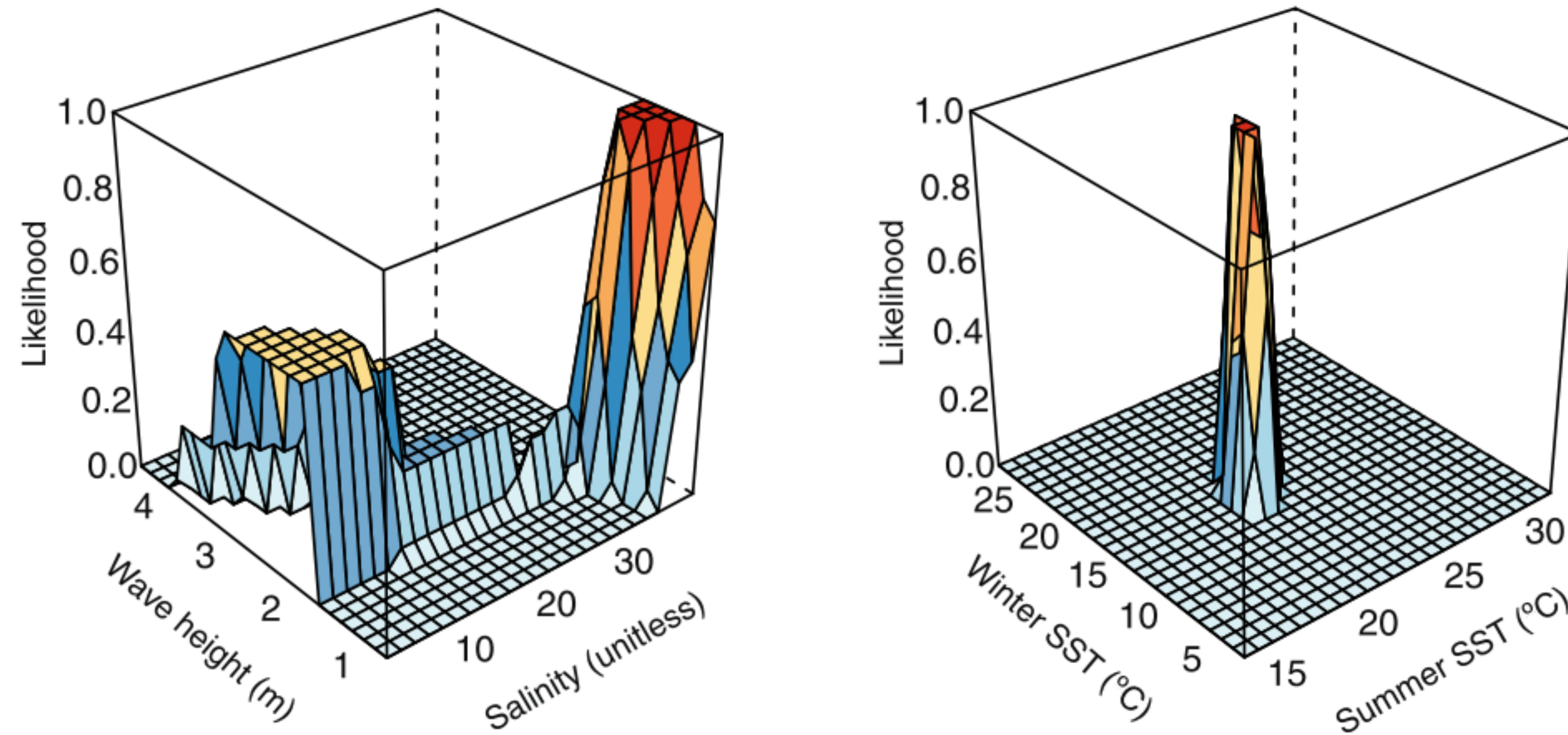
Hutchinson also distinguished between:

Fundamental niche: the **full range of environmental conditions and resources** (dimensions in the hypervolume) within which a species could potentially survive and reproduce indefinitely, **in the absence of biotic interactions**.

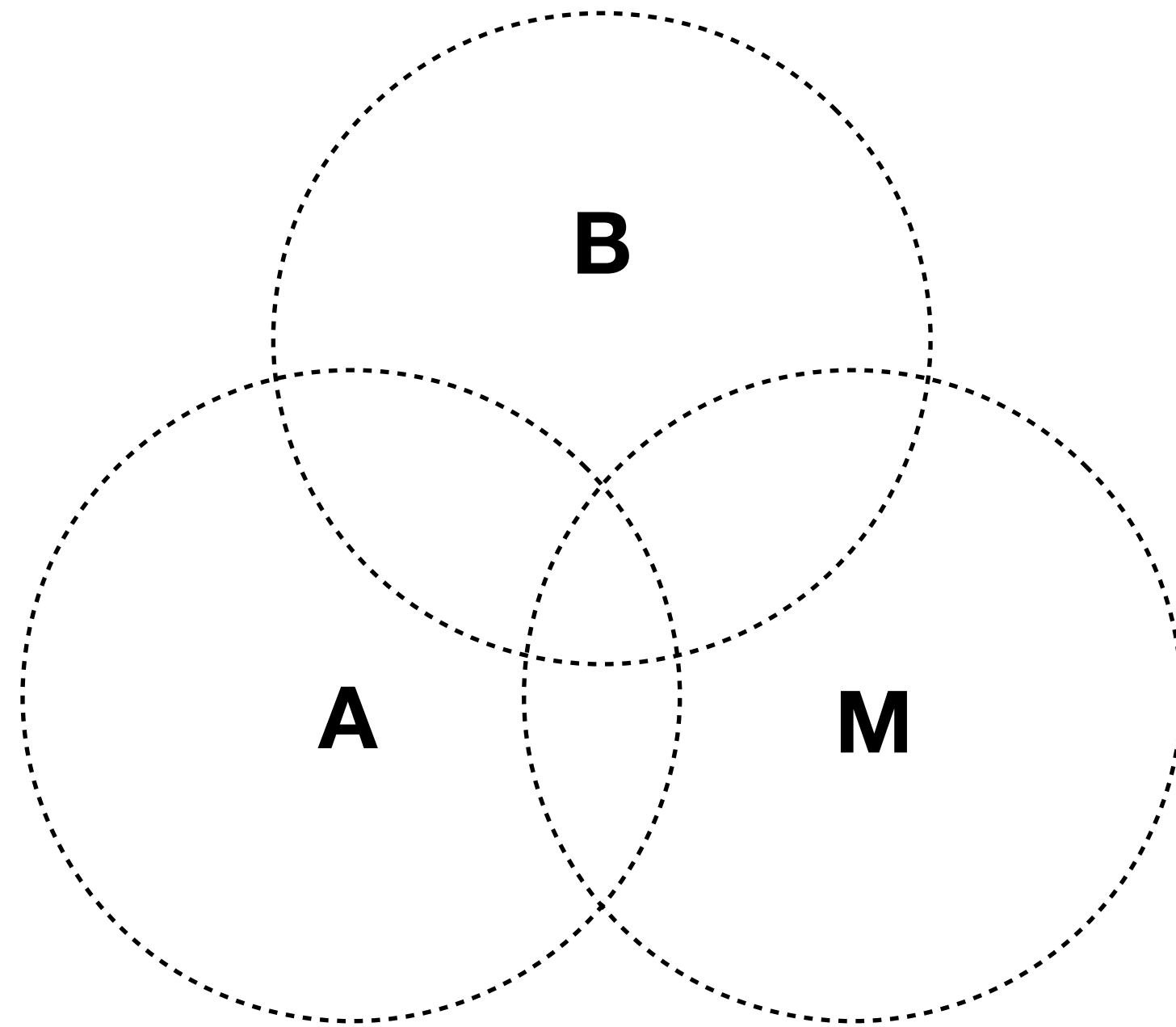
Realized niche: the **actual portion of the fundamental niche** (a subset of the hypervolume dimensions) **that a species actually occupies in the presence of limiting biotic interactions** - competition with other species for resources, or predation pressure restricts a species from utilizing its full fundamental niche.

The fundamental niche is the theoretical maximum niche space the species can occupy.

Realized Niche $\leq$ Fundamental Niche.



Hypervolume of abiotic conditions (environmental dimensions) where a Mediterranean coral can survive and reproduce (fundamental niche).



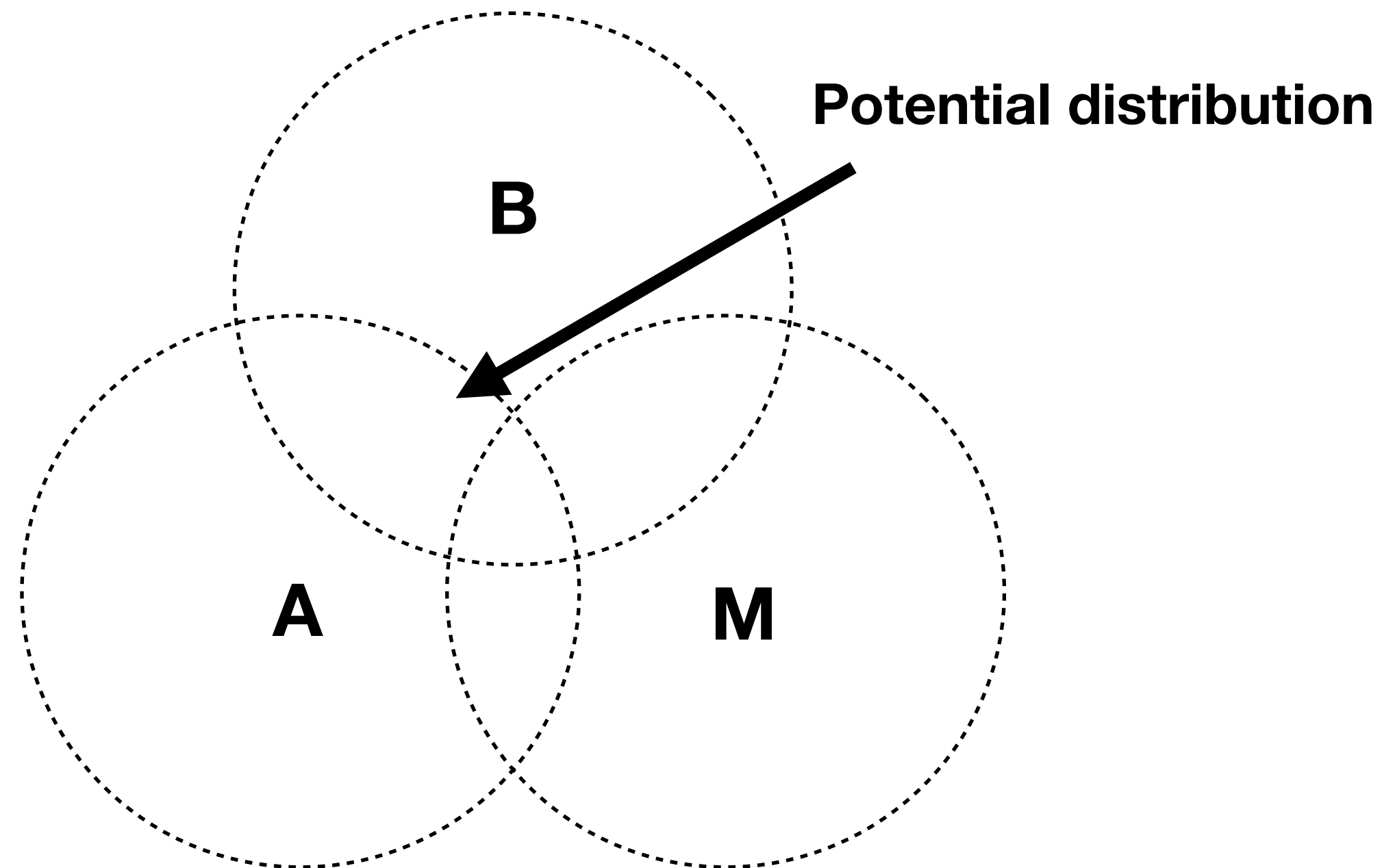
Connecting niche to geographic space: BAM diagram

A conceptual model integrating factors limiting distributions:

Biotic: suitable biotic conditions (presence of prey, absence of strong competitors).

Abiotic: suitable environmental conditions (temperature, salinity, depth, etc.).

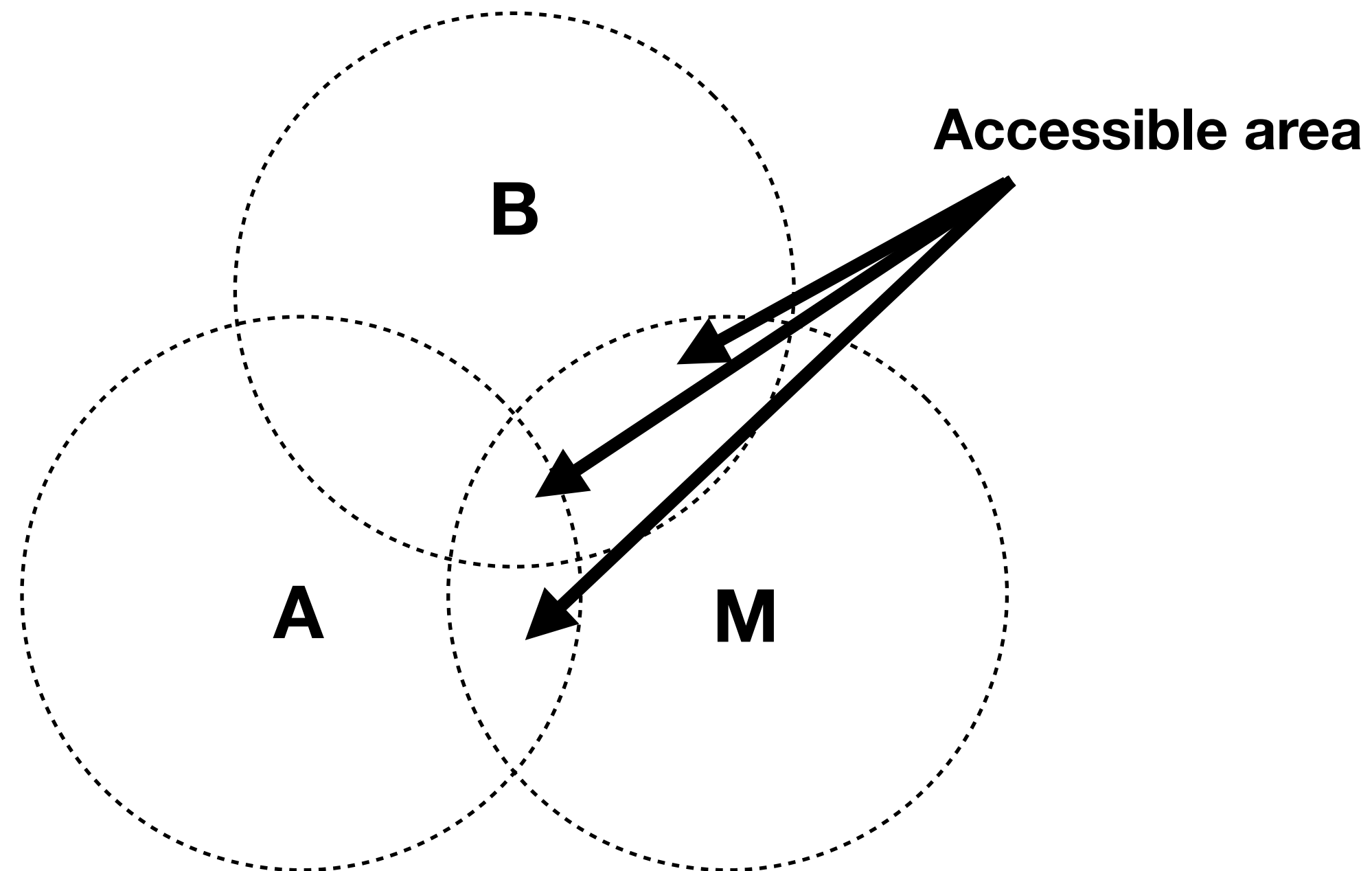
Movement: accessibility – the regions reachable by the species via dispersal.



Potential distribution ($A \cap B$): The geographic area where both **abiotic (A)** and **biotic (B)** conditions are **suitable**. The geographic expression of the **realized niche**.

Accessible area (M): This is the **geographic area the species can actually reach through dispersal** within a given time frame.

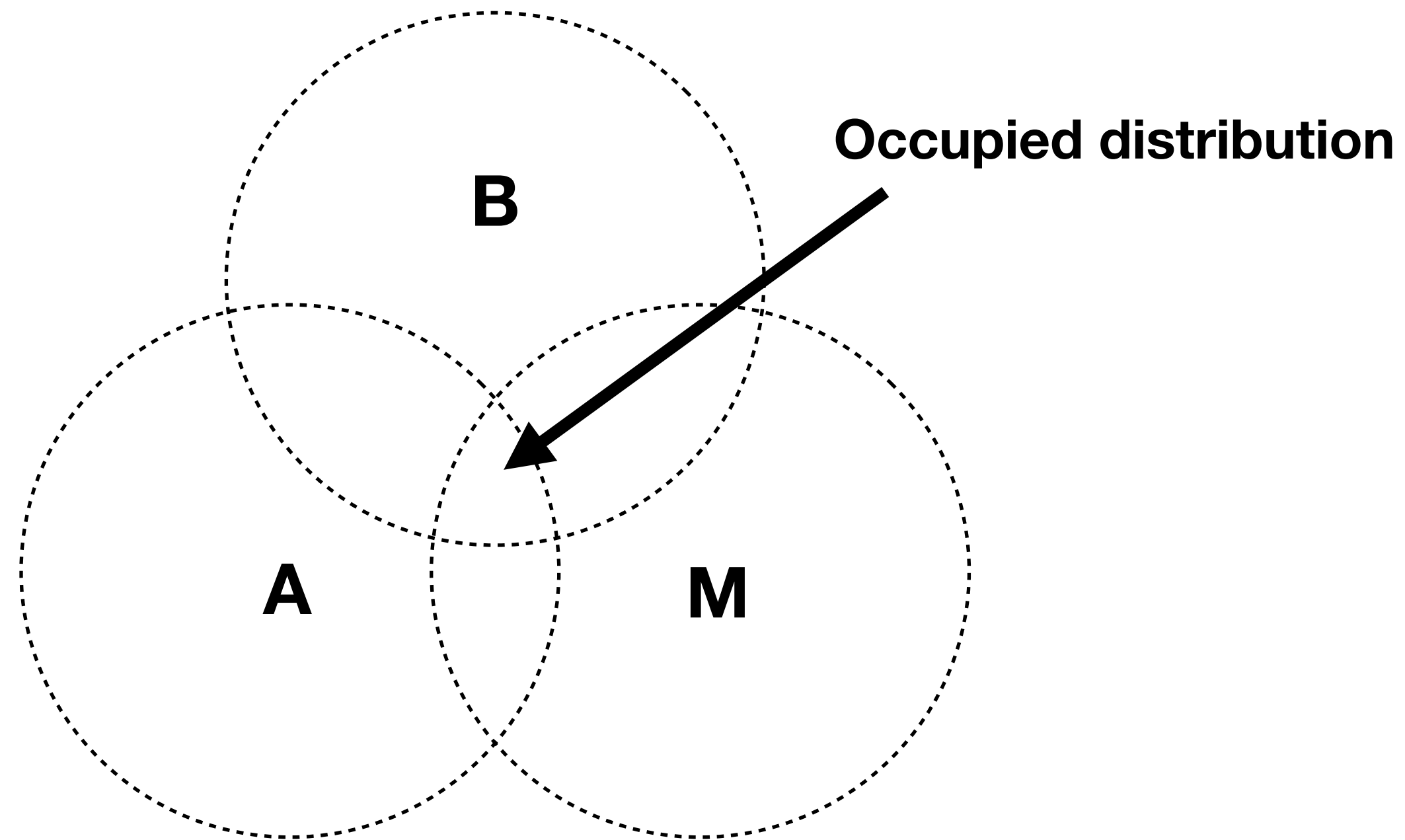
Occupied distribution ($A \cap B \cap M$): The **geographic area where the species is actually found**. It's the part of the potential distribution that is also accessible.



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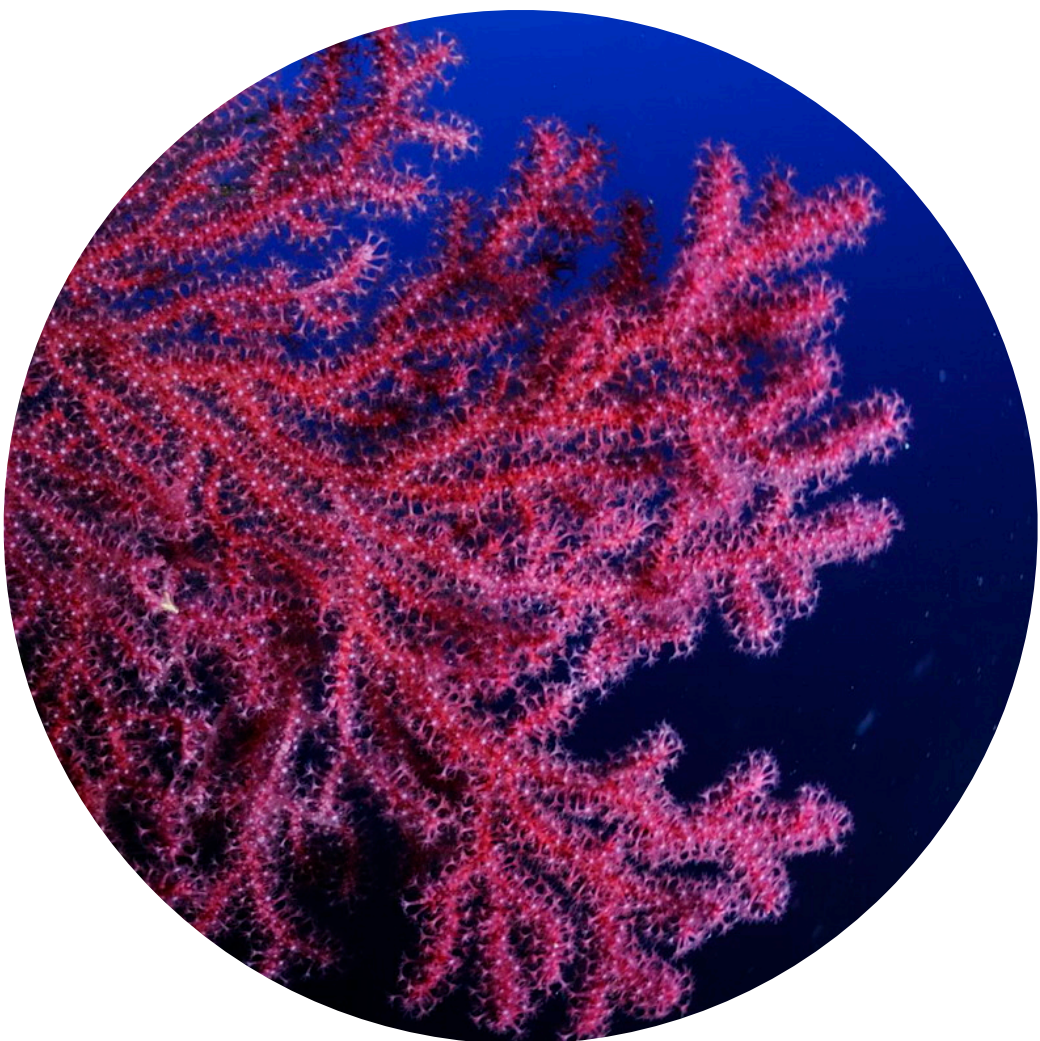
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Common marine limiting abiotic factors

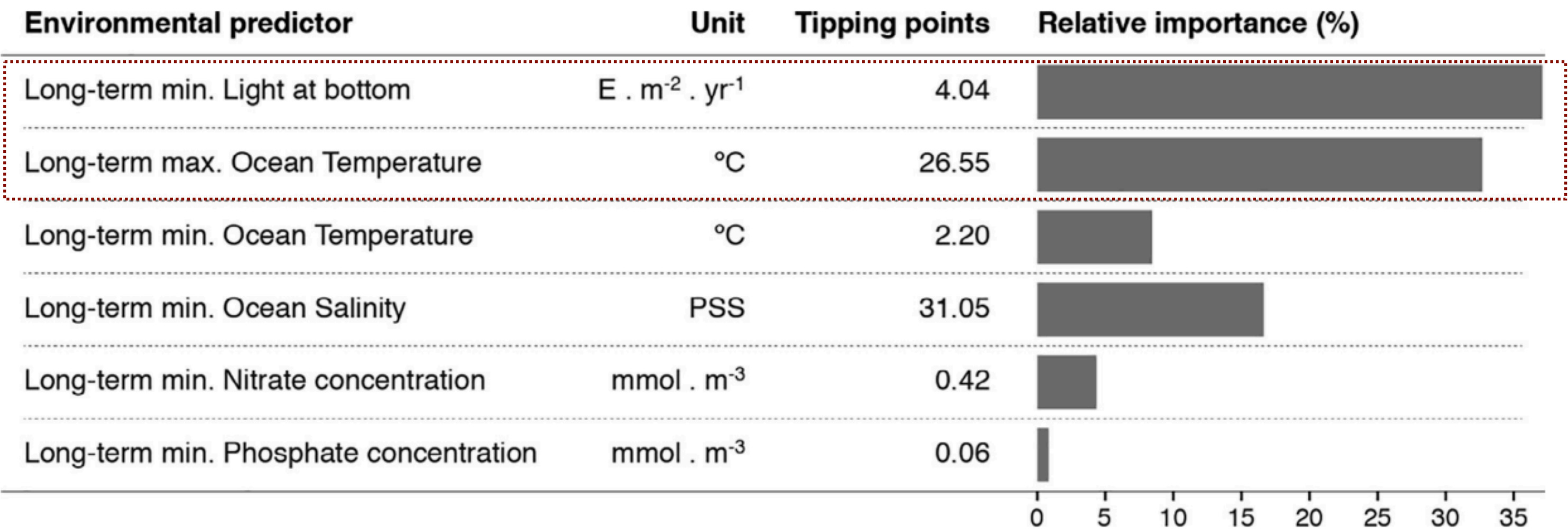
Mediterranean coral

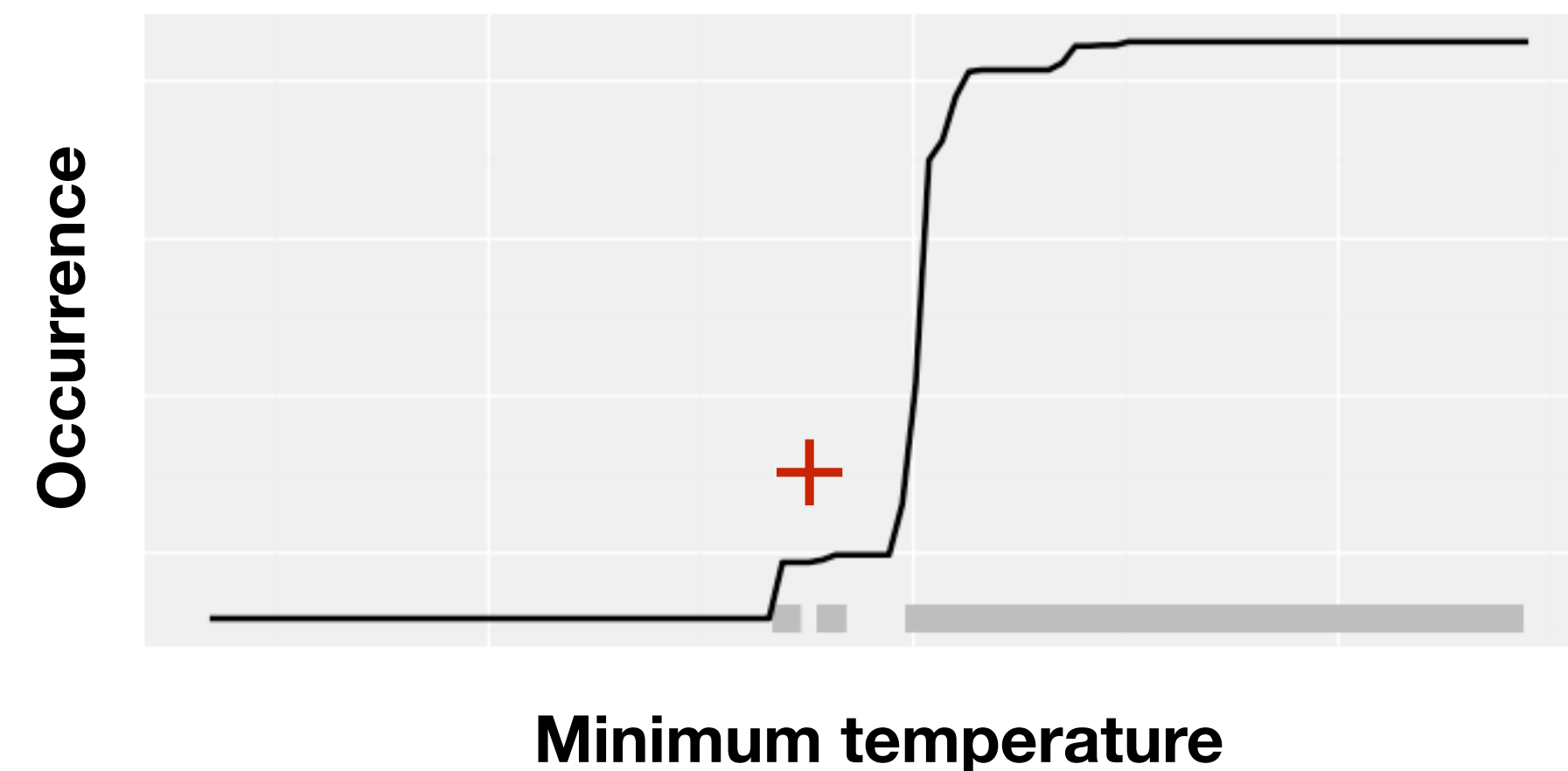
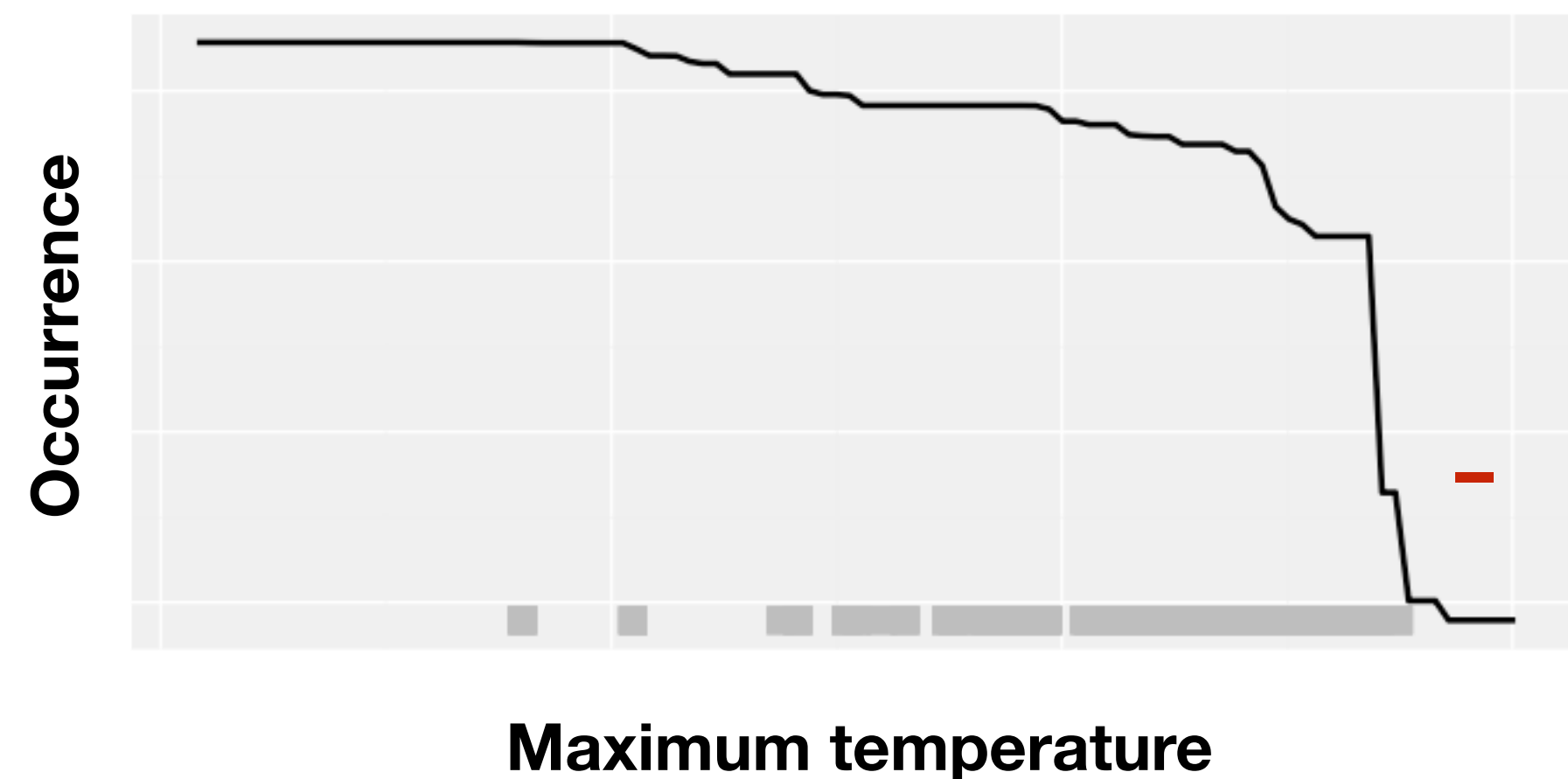
Predictor (unit)	Oc. range	LP model	LP literature	Relative contribution(%)
Temperature min. (°C)	11.4 - 16.0	<11.5	<12.0 ^{1,2}	
Temperature max. (°C)	14.1 - 24.4	>25.5	>25.0 ^{1,3,4}	
Slope (degree)	3.2e ⁻² - 22.2	<0.1	steep ¹	
Silicate max. (μmol/L)	1.6 - 19.9	>20.1		
Productivity min. (gC/m ³ /day)	7.5e ⁻⁸ - 3.3e ⁻³	<4.0e ⁻⁸		
Phosphate min. (μmol/L)	1.6e ⁻⁴ - 0.5	<2.2e ⁻⁵	< 0.08 ⁵	
Phosphate max. (μmol/L)	2.4e ⁻² - 0.77	>0.6		
Nitrate min. (μmol/L)	1.0e ⁻⁶ - 5.2	<7.2e ⁻⁷	< 2.0 ⁵	
Nitrate max. (μmol/L)	2.5e ⁻³ - 10.9	>5.4		



Common marine limiting abiotic factors

Agressive invasive macroalgae





Visualizing suitability: species response functions

How a species' suitability changes along an environmental gradient. Common shapes:

Unimodal: optimal range with declines towards low and high values (e.g., average temperature).

Monotonic: increasing or decreasing suitability (e.g., increased tolerance up to an extreme point - physiological theory). **Often skewed, reflecting stress near tolerance limits.**

Are niches fixed? Conservatism vs. evolution

A central question in Macroecology: How stable are species' ecological niches?

Two opposing perspectives:

Niche conservatism: niches remain stable over time.

Niche evolution: niches can evolve over time.

The balance between these perspectives influences species' distribution, their local adaptation, and speciation.

Niche conservatism

The tendency for **species to retain ancestral niche characteristics** over long periods (especially environmental tolerances).

Implications:

Implies **no adaptive or evolution potential** to novel conditions.

Species **respond to climate change by shifting their geographic distribution** (tracking suitable conditions) rather than adapting in place.

Evidence: observed distribution shifts match climate shifts. Without niche conservatism, species would persist while climate conditions shift.

Niche evolution

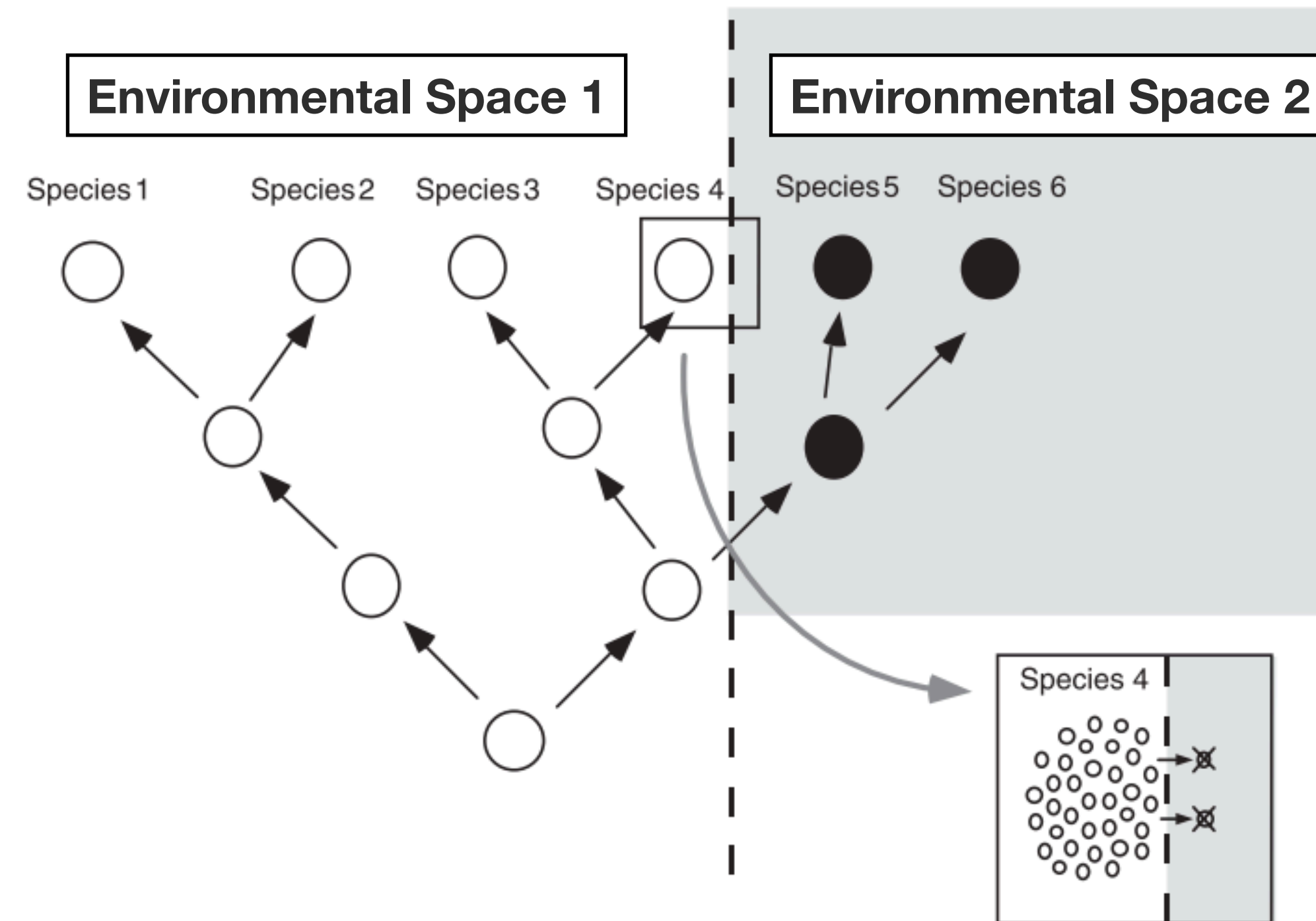
Niches can evolve, allowing species to colonize new environments.

Happens through natural selection acting on physiological tolerances, resource use, etc.

The key debate today: is niche evolution fast enough to keep pace with rapid, human-caused environmental change (e.g., climate change)?

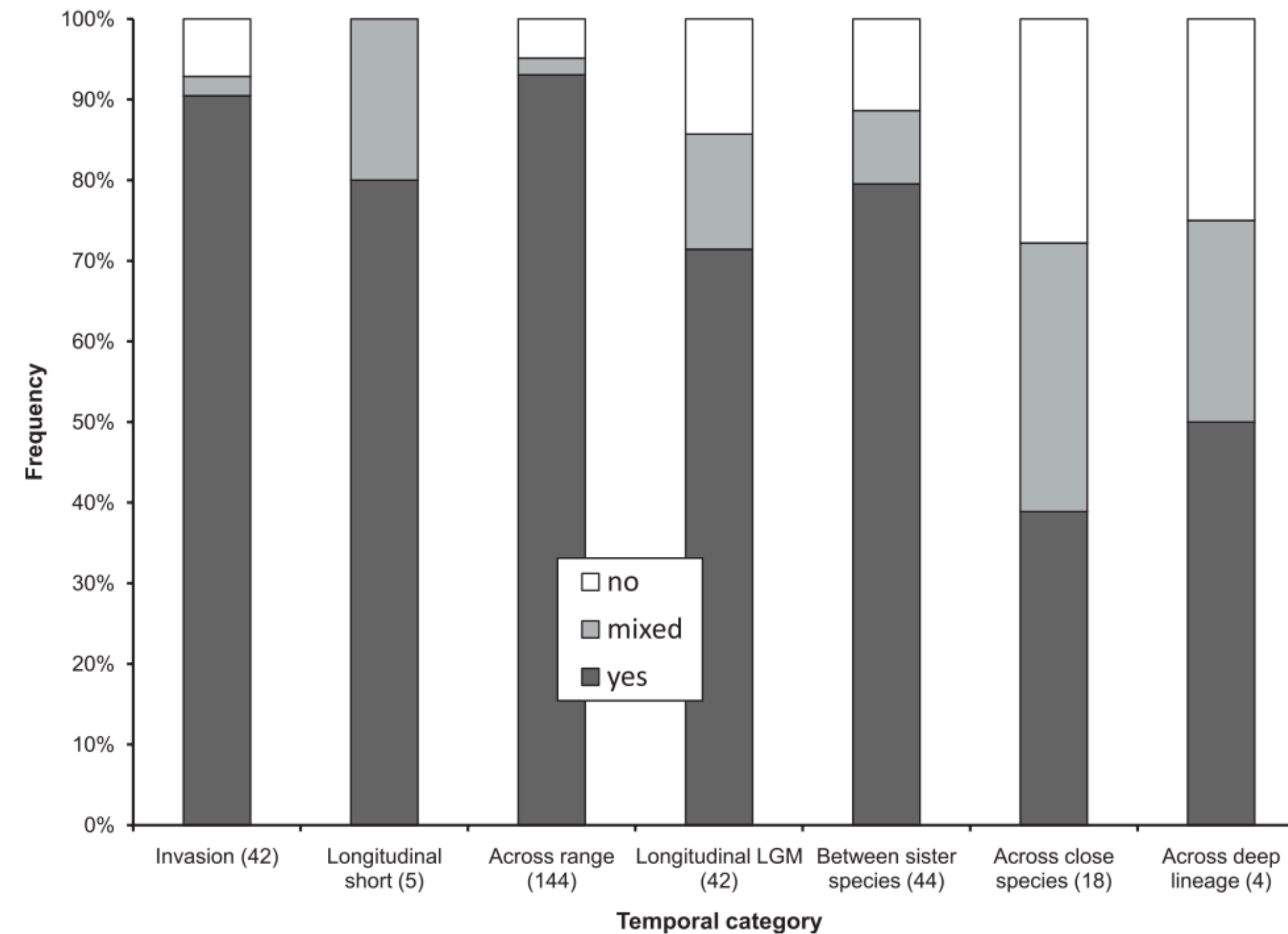
If evolution is slow relative to climate change, niche conservatism dominates responses (range shifts, local extinctions).

If fast enough, adaptation is possible under ongoing climate change.



In the past, for a given environmental space, **a clade originates from an ancestral species (niche evolution to a different conditions** - by genetic mutation and posterior selection).

At present, there are 4 species using the ES1 and 2 species using ES2. Niche conservatism in species 1-4 limits the individuals of using ES2, and vice-versa.



An extensive review **shows evidence for ecological niches being highly conserved over short-to-deep time spans.**

(i.e., from individual life spans - invasions - up to hundreds of thousands of years).

Even at deep geological time scales of thousands of years NC of ~50% of species.

References

Spalding et al., 2007: <https://doi.org/10.1641/B570707>

Fragkopoulou et al, 2022: <https://doi.org/10.1111/geb.13450>

Boavida et al, 2016: <https://doi.org/10.1038/srep36460>

Soberón & Peterson, 2005: <https://doi.org/10.17161/bi.v2i0.4>

Koerich et al., 2020: <https://doi.org/10.1016/j.scitotenv.2019.134692>

Wiens et al., 2010: <https://doi.org/10.1111/j.1461-0248.2010.01515.x>

Peterson et al., 2011: <https://doi.org/10.1111/j.1365-2699.2010.02456.x>